

Yang Sen (Liam) Lin

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Education

National Yang Ming Chiao Tung University (NYCU) Sep 2021 – Dec 2025 (expected)
B.S., Mechanical Engineering & Computer Science (Cross-disciplinary) GPA: **4.08/4.3** (last 60), 3.85/4.3 (overall)
Academic Excellence Award × 2 (**Top 5% of Dept.**)
Ranking: **7/61 (11.4%)**

Relevant Coursework (Control, Math & Systems): Automatic Control I & II (**A+/A**), Engineering Math I & II (**A-**), Applied Electrics (**A+**), Smart Materials (**A+**), Applied Mechanics (Dynamics) (**A**).

Relevant Coursework (AI, Vision & Algorithms): Computer Vision (Graduate Level) (**A+**), Intro to Algorithms (**A**), Intro to AI (**A+**), CV for UAV Autopilot (**A**), Data Structures & OOP (**A**).

Publications

[1] See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation. **CoRL 2025 (Accepted)**

Chih Yao Hu*, **Yang-Sen Lin***, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, et al. (* Equal Contribution)

(Note: CoRL ranked **#3 in Robotics** on Google Scholar, **H5-index: 107**)

Research Impact: Proposed a **zero-shot** navigation framework that decouples VLM reasoning from geometric control, achieving **SOTA** success rates in unstructured environments without task-specific training.

Technical Novelty: Formulated a **geometric projection algorithm** to ground 2D visual prompts into **3D metric controls**, solving the **depth ambiguity** problem inherently found in monocular vision setups.

Work & Research Experience

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Incoming Research Intern — Mentor: Dr. Masashi Hamaya

Tokyo, Japan

Jun. 2026 – Aug. 2026

(Selected for a research position focusing on Robotics Foundation Models)

- **Proposed Research:** To formulate a knowledge distillation framework to compress large-scale Vision-Language-Action (VLA) models, aiming to resolve computational bottlenecks for **real-time inference** on **edge hardware**.
- **Research Goal:** To investigate **latency minimization** techniques for **high-frequency control loops** in soft robotics, with a target of submitting findings to top-tier conferences (e.g., **CoRL/ICRA**).

Computational Photography Lab, NYCU

Undergraduate Researcher — Advisor: Prof. Yu-Lun Liu

Hsinchu, Taiwan

Sep. 2024 – Present

- **Geometric Formulation (The Math):** Derived a **mathematical action-to-control mapping** to ground high-level VLM inference (2D coordinates) into low-level drone dynamics (3D trajectories), theoretically solving depth ambiguity without depth sensors.
- **Sim2Real Engineering (The Validation):** Overcame the simulation-to-reality gap by calibrating coordinate systems and **optimizing control latency**. Validated robustness through **55+ real-world flight experiments**, demonstrating long-horizon planning in unstructured environments.

Wolley Inc.

Firmware Engineer Intern — Advisor: Dr. Yu-Ming Chang (Director)

Hsinchu, Taiwan

Jul. 2024 – Aug. 2024

- **System Profiling (CXL/Memory):** Engineered C-based diagnostic tools to profile **throughput and latency** for **CXL Type-3** solutions, providing critical metrics to reduce **host bandwidth pressure**.
- **Parallel Computing & Optimization:** Refactored the internal testing system using **parallel computing techniques** in Linux (optimizing I/O efficiency), which significantly reduced unit testing time for the main product line.

Technical Skills

- **Languages:** Python (*Proficient*), C/C++ (*System-level & Embedded*), C# (*Unity*).
- **AI & Computer Vision:** OpenCV, YOLO, **VLM Integration**, PyTorch, GANs.
- **Robotics & Control:** PID Control, UAV Dynamics, SDL (*Physics simulation*).
- **Systems & Tools:** Linux (*Profiling*), Git, Docker, **CXL/HDM Protocols**, Embedded Profiling Tools.

Selected Projects

Vision-Based UAV Autopilot System | *Python, OpenCV, Control Theory* 🤖

- **Algorithmic Optimization:** Optimized visual tracking algorithms from **quadratic $\mathcal{O}(N^2)$ to linear $\mathcal{O}(N)$ complexity** via **heuristic-based sparse sampling**, eliminating computational bottlenecks for real-time control.
- **Control Implementation:** Engineered a **closed-loop PID controller** from scratch, iteratively tuning proportional and derivative gains (K_p, K_d) to achieve **<3cm hover accuracy** without relying on external positioning systems.

Perceptual Image Restoration via High-Order Modeling | *Python, PyTorch, Signal Processing* 📷

- **Mathematical Modeling:** Modeled signal degradation via **Generalized Gaussian kernels** ($D = (k \otimes I)_{\downarrow s} + n$) to simulate complex blur artifacts.
- **Perceptual Alignment:** Mitigated domain shift by fine-tuning generative priors, achieving superior **Mean Opinion Scores (MOS)** in perceptual fidelity compared to standard interpolation baselines.

DesignMate: Computational AR Sketching Interface | *Unity, C#, Linear Algebra* 🎨

- **Geometry Pipeline:** Developed a **vector-to-mesh pipeline** using **cubic interpolation** to smooth discrete stylus inputs, driving a diffusion-based generation workflow.
- **Resource Efficiency:** Designed a robust polling mechanism to manage **GPU memory allocation**, preventing bottlenecks on **mobile AR hardware** during distributed inference.

Leadership & Service

Google Developer Student Club (GDSC) NYCU

Chapter Lead

Hsinchu, Taiwan

Jul. 2024 – Jun. 2025

- **Organizational Leadership:** Directed a 24-member team across 5 departments; supervised **6 AI & Software Engineering projects** bridging academic theory with industry practice.
- **Community Impact:** Orchestrated 10+ technical workshops, impacting **40+ student developers** and fostering a peer-learning ecosystem.

Meichu Hackathon (Logitech Division)

Tech Lead & PM (3rd Place Winner)

Hsinchu, Taiwan

Sep. 2025

- **Technical Management:** Led a cross-functional team of 5 to develop *DesignMate* under a **48-hour deadline**; **architected the system framework** based on team expertise.
- **Communication:** Delivered the final technical pitch, successfully communicating complex engineering concepts to non-technical judges.

NYCU Varsity Swim Team

Team Member & Relay Gold Medalist

Hsinchu, Taiwan

Jan. 2022 – Dec. 2024

- **Achievement of Grit:** Progressed from novice to **City-Level Gold Medalist** within 3 years. Balanced **8+ hours of weekly high-intensity training** with rigorous research commitments.

TEDxNYCU

Selected Speaker

Hsinchu, Taiwan

Dec. 2023

- **Community Leadership:** Selected to address an audience of 50+ on "Redefining Success"; utilized a structured narrative framework to articulate personal value systems, inspiring purpose-driven learning among university peers.

Community Service: Homeless Support Volunteer

Initiator & Logistics Lead

Taipei, Taiwan

Sep. 2025 – Nov. 2025

- **Social Initiative:** Initiated a 10-week meal distribution program; managed procurement logistics and led weekly reflection sessions to improve service delivery efficiency.

Honors & Awards

- **3rd Place**, Meichu Hackathon (Logitech Division) *Sep. 2025*
- **2nd Place & Popularity Award**, Taipei Metro Hackathon (*Ranked 2nd out of 92 teams*) *May 2024*
- **Gold Medal**, Men's 4x100m Relay, Hsinchu City Swimming Championship *Sep. 2024*
- **Academic Excellence Award** × 2 (*Top 5% of Dept.*) *Fall 2023, Fall 2024*